Project Specification: AI Environmental Impact Tracker

Overview

The goal of this project is to build an AI-powered environmental impact tracker that enables anyone to calculate, understand, and reduce their personal environmental footprint.

The application should be free, easy to use, and designed for the general public, with a particular focus on South African and broader African households. A user with no environmental knowledge should be able to complete an assessment in less than 10 minutes while still receiving a realistic estimate of their impact.

The application should reflect local realities such as municipal utilities, prepaid meters, loadshedding, backup generators, solar inverters, mixed household energy sources, and regional differences across South Africa and Africa.

The application measures three primary environmental metrics:

- Net Water Consumption
- Net Electricity Consumption
- Net Carbon Footprint (CO_2e)

The application should prioritize real measured data, such as utility bills or meter statements, whenever possible, and only estimate values when no measured data is available.

The objective is not only to calculate a footprint, but also to educate users by explaining where their environmental impact comes from and providing practical, personalized recommendations for reducing it.

Core Principles

The application should follow these principles throughout its design.

Simplicity

Users should answer as few questions as possible.

Questions should only be asked if they significantly improve the accuracy of the calculations.

Whenever information can reasonably be inferred or calculated from existing answers, the application should avoid asking for additional questions.

Accuracy

Whenever possible, calculations should be based on measured data instead of estimates.

For example:

- Utility bills or prepaid meter statements are preferred over behavioural estimates.
- Vehicle databases are preferred over asking users for fuel efficiency.
- Airport databases should be used instead of asking users to estimate flight distances.
- Regional electricity and water datasets should be used where available for South African and African contexts.

Accessibility

The application should be suitable for people who know very little about environmental sustainability.

Questions should be written in simple language.

Where necessary, small explanations should help users understand why information is being requested.

Transparency

Users should be able to understand how their footprint is calculated.

The application should clearly identify the activities contributing most to their environmental impact.

User Flow

The overall user journey should be:

1. Create an account or sign in.
2. Choose an assessment method.
3. Provide environmental data.
4. Calculate environmental footprint.
5. View dashboard.
6. Receive AI-generated analysis.
7. Save results for future comparisons.

Two Assessment Methods

The application should allow users to complete the assessment in one of two ways.

Method 1 — Guided Questionnaire

The application asks structured questions using forms, sliders, dropdowns, checkboxes and file uploads.

This method is intended for users who prefer a traditional questionnaire.

Method 2 — AI Conversation

The user chats naturally with an AI assistant.

The AI should ask follow-up questions until it has collected all required information.

The chatbot should ultimately collect the exact same information required by the questionnaire.

Both assessment methods should produce identical calculations and identical final results.

Information Collected

General Information

Collect:

- Household size
- Country
- Province, state, or region (optional)
- Municipality or local utility (optional)

This information allows the application to use regional emission factors and environmental datasets relevant to South Africa and other African countries.

Water Assessment

The application should always prioritize actual measured water consumption.

Preferred Option

Users should be able to upload a municipal water bill or water statement.

If uploaded, the application should extract:

- Monthly water consumption
- Water bill amount

Manual Entry

If no bill is available, users should be able to enter:

- Monthly water usage

or

- Monthly water bill

Rainwater Harvesting

Ask:

Does your household use rainwater harvesting?

If yes:

Approximately what percentage of your household's water comes from rainwater?

Rainwater should be treated as having a significantly lower environmental impact than municipally supplied water.

Estimation Questions

Only ask these questions if no reliable water usage information is available.

- Average shower duration
- Number of showers per week
- Garden irrigation
- Swimming pool ownership

These questions should only be used to estimate water consumption when measured data is unavailable.

Electricity Assessment

The application should prioritize measured electricity consumption.

Preferred Option

Allow users to upload an electricity bill or prepaid meter statement.

Extract:

- Monthly electricity consumption

- Electricity bill amount

Manual Entry

If no bill is available:

Allow users to enter:

- Monthly electricity consumption

or

- Monthly electricity bill

Renewable Electricity

Ask:

Does your household generate renewable electricity?

Options include:

- Solar
- Wind
- None

If renewable electricity is used:

Estimate the percentage of household electricity supplied by renewable sources.

Backup Power and Loadshedding

Because loadshedding is a relevant part of daily life in South Africa and some other African contexts, ask whether the household uses backup power during outages.

Options include:

- Generator
- Inverter and battery
- Solar battery backup
- UPS
- None

If a generator is used, collect:

- Fuel type (petrol, diesel, gas, or other)
- Average hours used per month or litres of fuel used per month

- Whether it powers the whole home or only essential loads

If inverter, battery, or solar backup is used, collect:

- Approximate share of household electricity supplied during outages
- Whether the system is charged from the grid, solar, or both

Backup power should be included in the carbon footprint calculation, and where relevant, in the overall electricity-related analysis.

Estimation Questions

Only if measured electricity usage is unavailable:

Collect:

- Home size
- Electric geyser
- Air conditioner
- Electric stove
- Pool pump

These values should estimate household electricity consumption.

Carbon Footprint Assessment

The carbon footprint should include transport, household energy, lifestyle, consumption, and backup power used during loadshedding.

Flights

Allow users to add multiple flights.

For each flight collect:

- Departure airport
- Arrival airport
- Cabin class
- One-way or return flight
- Number of trips per year

Flight distances should be calculated automatically using airport locations.

Personal Vehicle

Collect:

- Vehicle manufacturer
- Vehicle model
- Model year
- Annual distance driven
- Average number of passengers

The application should determine fuel type and fuel economy automatically whenever possible.

Public Transport

Collect:

- Transport type
- Average distance travelled each week

Supported transport types include:

- Bus
- Train
- Minibus taxi / taxi
- Subway
- Other

Home Heating

Collect:

Primary heating method.

Examples:

- Electricity
- Gas
- Coal
- Wood
- Paraffin / kerosene
- Heat pump
- None

Also collect:

- Months used annually
- Average hours used per day

Diet

Allow users to choose the description that best matches their diet.

Options:

- Heavy meat eater
- Average diet
- Mostly chicken
- Pescatarian
- Vegetarian
- Vegan

Food Waste

Estimate the percentage of purchased food that is wasted.

Options include:

- None
- Very little
- Around 10%
- Around 20%
- More than 30%

Shopping Habits

Ask users to compare their purchasing habits with the average person.

Options:

- Much less
- Less
- Average
- More
- Much more

This should represent purchases such as clothing, electronics and household goods.

Carbon Offsets

Ask whether users purchase certified carbon offsets.

If yes:

Collect the approximate annual amount of carbon offset.

Calculation Philosophy

The application should calculate environmental impact using established environmental conversion factors and scientific datasets.

The calculations should be deterministic and reproducible.

Where measured data exists, it should always take priority over estimated data.

Behavioural estimates should only be used when measured values are unavailable.

Where possible, the application should use South African and African regional datasets, including local electricity grid factors, municipal water data, transport assumptions, and country-specific emission factors.

Artificial Intelligence

Artificial intelligence serves two roles within the application.

1. Conversational Data Collection

Users should be able to complete the assessment naturally through conversation.

The AI should ask follow-up questions until all required information has been collected.

The chatbot should extract structured information from natural language responses.

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2. Environmental Analysis

Once calculations have been completed, AI should analyse the calculated results.

The AI should:

- Explain the user's environmental footprint.
- Identify the largest contributors.
- Explain why those contributors matter.
- Recommend practical improvements.
- Prioritize recommendations based on impact and feasibility.
- Estimate the environmental benefit of each recommendation where possible.

The AI should function as an environmental sustainability advisor rather than as a calculator.

Dashboard

After completing the assessment, users should receive a dashboard summarizing their results.

The dashboard should include:

- Total water consumption
- Total electricity consumption
- Total carbon footprint

It should also provide category breakdowns, identify the user's largest contributors, compare results against South African, African, national, and global averages where available, and clearly show opportunities for improvement.

Long-Term Vision

The application should become a personal environmental companion rather than a once-off calculator.

Users should be encouraged to repeat assessments over time and monitor their progress.

Future versions may include:

- Historical tracking
- Goal setting
- Gamification
- Community challenges
- Sustainability achievements and badges
- Integration with smart meters, prepaid meters, utility providers, solar inverters, and backup power systems
- Personalized progress reports
- More detailed regional environmental datasets across South Africa and Africa

The overall vision is to create an accessible, scientifically grounded platform that empowers individuals to understand and reduce their environmental impact through accurate measurement, education, and actionable AI-driven guidance.